

IoT Applications in Smart Agriculture

Research Report | Internet of Things (IoT) – Task 1

Introduction

Agriculture is affected by changing weather, limited water availability, pests, and the difficulty of making timely decisions in the field. The Internet of Things (IoT) provides a way to address some of these challenges by connecting sensors, controllers, communication networks, and actuators. These technologies allow agricultural conditions to be measured continuously and provide information that can support better decisions. This approach is commonly associated with Smart Agriculture or Precision Farming, where field-level data is used alongside the farmer's experience [1], [2].

1. What Makes an Agricultural System Smart

An IoT-based agricultural system can be understood as a four-stage process: sense, transmit, analyze, and act. Sensors can measure soil moisture, soil pH, temperature, humidity, and sunlight intensity. The collected readings may be transmitted through technologies such as Wi-Fi, LoRa, Zigbee, or cellular networks such as NB-IoT, either to a local gateway or to a cloud platform [1], [4]. Software can then process the information using threshold-based rules and, in more advanced systems, machine-learning techniques. The result may be a notification to the farmer or an automatic action such as switching on a water pump [1], [3].

2. Major IoT Applications in Agriculture

Smart Irrigation: Smart irrigation is an important IoT application in agriculture, where soil-moisture sensors can be used to support irrigation decisions [1], [2]. A sensor placed near the crop root zone can continuously provide information about soil moisture. When the measured value falls below a selected threshold, a controller can activate a pump or solenoid valve. The irrigation system can then stop after the desired moisture condition is reached. This approach differs from fixed schedules because watering is linked to actual field conditions [1], [3].

Environmental and Crop Health Monitoring: IoT sensor nodes can monitor environmental parameters such as temperature, humidity, rainfall, and wind speed. These measurements provide field-level information that can help farmers observe changing conditions. Some agricultural systems also use cameras, including multispectral or infrared cameras, to identify possible crop stress, nutrient deficiency, or disease symptoms at an early stage [1], [3].

Livestock Monitoring: IoT applications can also be used for livestock management. Wearable devices or tags can provide information about animal location, temperature, and activity. Such monitoring can help farmers observe changes in animal behaviour and locate livestock across large or remote areas [3], [5].

Drone-Based Field Surveys: Agricultural drones equipped with cameras and sensors can survey large areas and generate information about crop growth, waterlogging, and affected patches. This can support targeted application of fertilizers or pesticides instead of treating the entire field uniformly [1], [6].

Storage and Supply Chain Monitoring: IoT can continue to be useful after crops leave the field. Sensors installed in grain storage or cold-storage environments can monitor temperature and humidity to help reduce spoilage. GPS-based tracking can also provide information about the movement of agricultural products during transportation [3], [6].

3. Practical Example of a Smart Irrigation System

A small-scale smart irrigation system can be built using a soil-moisture sensor, a DHT22 temperature-humidity sensor, and an ESP32 microcontroller. The ESP32 can read sensor values periodically and send them to a cloud platform using the MQTT protocol. A dashboard made with a platform such as Node-RED or Blynk can display current readings and historical trends. If soil moisture falls below a chosen level, the ESP32 can control a relay connected to a water pump. The farmer can also receive an application or SMS notification. This type of setup uses commonly available components and demonstrates how sensing, communication, data processing, and actuation can be combined in one IoT system [1], [3], [4].

4. Challenges in Adoption

Although IoT offers several possibilities for agriculture, its adoption has practical challenges. Poor internet connectivity in rural areas can make cloud-based systems difficult to use, while low-power wide-area technologies such as LoRaWAN can provide an alternative in suitable locations. The initial cost of sensors, gateways, communication services, and other equipment may also be difficult for small and marginal farmers. Outdoor sensors require maintenance because they are exposed to weather, dust, and animals. In addition, users may need training and local technical support to install and troubleshoot IoT systems. Data privacy is another consideration because farm information may contain details that farmers do not want to share with third parties [1], [3].

5. Conclusion

IoT is helping agriculture move toward greater use of real-time, field-level information. Applications such as smart irrigation, environmental monitoring, livestock tracking, drone surveys, and storage monitoring show how sensors and connected systems can support agricultural operations [1], [2], [3]. A basic system can be created from sensors, a microcontroller, communication technology, and suitable control devices, making the concept applicable beyond only large commercial farms. However, connectivity, cost, maintenance, technical training, and data privacy remain important factors in wider adoption. Addressing these practical issues will be important for the continued development of smart agriculture [1], [3], [5].

References

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